



Air University
(Final-Term Examination: Spring 2025)

Subject: **Object-Oriented Programming**
Course Code: **CS-112**
Class: **BS-CYS**
Semester: **IV**
Section: **Evening A/B**

Total Marks: **100**
Date:
Time:
Duration: **3 Hours**
FM Name: **Khwaja Mansoor**

HoD Signatures: _____

FM Signatures: _____

Note:

- All questions must be attempted.
- Answers must be precise and to the point focus on writing clean, well-structured code and providing a clear rationale for your implementation rather than simply restating concepts from the question.

Q. No. 1 (CLO 2)		60 Marks
Dry Run		
a	<pre>#include <iostream> #include <string> using namespace std; class Student { private: string name; int marks[3]; public: Student(string n, int m1, int m2, int m3) { name = n; marks[0] = m1; marks[1] = m2; marks[2] = m3; } int totalMarks() { return marks[0] + marks[1] + marks[2]; } float averageMarks() { return totalMarks() / 3.0; } void display() {</pre>	20

```

        cout << "Name: " << name << endl;
        cout << "Marks: ";
        for (int i = 0; i < 3; i++) {
            cout << marks[i] << " ";
        }
        cout << "\nTotal: " << totalMarks();
        cout << "\nAverage: " << averageMarks() << endl;
        cout << "-----\n";
    }

    string getName() {
        return name;
    }
};

int main() {
    Student students[3] = {
        Student("Ali", 85, 90, 88),
        Student("Sara", 78, 82, 80),
        Student("Zain", 92, 95, 89)
    };

    float highestAvg = 0.0;
    string topStudent = "";
    for (int i = 0; i < 3; i++) {
        students[i].display();
        if (students[i].averageMarks() > highestAvg) {
            highestAvg = students[i].averageMarks();
            topStudent = students[i].getName();
        }
    }

    cout << "Top Student: " << topStudent << " with average
marks: " << highestAvg << endl;
    return 0;
}

```

```

#include <iostream>
#include <cstring>
using namespace std;

```

```

class Student {
private:
    char* name;
    int* marks;
    int numSubjects;
    float average;
public:
    Student(const char* n, int m[], int count) {
        numSubjects = count;
        name = new char[strlen(n) + 1];
        strcpy(name, n);
        marks = new int[numSubjects];
        for (int i = 0; i < numSubjects; i++) {
            marks[i] = m[i];
        }
        calculateAverage();
    }
    Student(const Student& s) {
        numSubjects = s.numSubjects;
        name = new char[strlen(s.name) + 1];
        strcpy(name, s.name);
        marks = new int[numSubjects];
        for (int i = 0; i < numSubjects; i++) {
            marks[i] = s.marks[i];
        }
        average = s.average;
    }
    void calculateAverage() {
        int sum = 0;
        for (int i = 0; i < numSubjects; i++) {
            sum += marks[i];
        }
        average = static_cast<float>(sum) / numSubjects;
    }
    void changeName(const char* newName) {

```

```

        strcpy(name, newName);
    }

    void display() const {
        cout << "\nName: " << name << endl;
        cout << "Marks: ";
        for (int i = 0; i < numSubjects; i++) {
            cout << marks[i] << " ";
        }
        cout << "\nAverage Marks: " << average << endl;
    }

    ~Student() {
        delete[] name;
        delete[] marks;
    }
};

// Main function
int main() {
    char name[50];
    int m[3];
    cout << "Enter student name: ";
    cin >> name;
    cout << "Enter marks for 3 subjects:\n";
    for (int i = 0; i < 3; i++) {
        cout << "Subject " << i + 1 << ": ";
        cin >> m[i];
    }

    Student s1(name, m, 3);
    Student s2 = s1;
    cout << "\nBefore modifying s1:\n";
    s1.display();
    s2.display();
    s1.changeName("Zain");
    cout << "\nAfter changing s1's name:\n";
    s1.display();
    s2.display();
}

```


return 0;

}

Dry run the code with the following test case:

Enter student name: **Ali**

Enter marks for 3 subjects:

Subject 1: **80**

Subject 2: **75**

Subject 3: **90**

Analyze the following C++ program, which contains several syntax and logical errors. Identify and correct at least five errors. For each correction, underline the error and add a comment explaining the purpose of the correction. Ensure the corrected program compiles and produces the expected output shown at the end.

```
#include<iostream>
```

```
#include<string>
```

```
using namespace std;
```

```
class Student {
```

```
    string name;
```

```
    int marks[3];
```

```
    float average;
```

```
public
```

```
    void inputData() {
```

```
        cout << "Enter student name: ";
```

```
        cin >> name;
```

```
        cout << "Enter marks of 3 subjects: ";
```

```
        for (int i = 0; i <= 3; i++) {
```

```
            cin >> marks[i];
```

```
        }
```

```
    }
```

```
    void calculateAverage() {
```

```
        int sum = 0;
```

```
        for (int i = 0; i < 3; i++)
```

```
            sum = marks[i]; // Error here
```

```
            average = sum / 3;
```

```
    }
```

```
    void displayData() {
```

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```

        cout << "Student Name: " << name << endl;
        cout << "Average Marks = " << average << endl;
    }
};
int main() {
    Student s;
    s.inputData();
    s.calculateAverage();
    s.displayData();
    return 1;
}

```

User Input: Ali 80 75 90

Output:

Student Name: Ali

Average Marks = 81.6667

Q. No. 2 (CLO 2)

20 Marks

Write a Rectangle class with private member variables length and width (both of type double), a default constructor that initializes both to 0.0, a parameterized constructor to set them, mutator and accessor functions for both variables, and member functions to calculate and return the area (length * width) and perimeter (2 * (length + width)); then write a program that asks the user to input the length and width, creates a Rectangle object, and displays its area and perimeter.

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Q. No. 3 (CLO 3)

20 Marks

You are developing a basic Threat Detection System for a cybersecurity application. The system needs to handle multiple types of threats such as Malware, Phishing, and Ransomware. Each threat type has different characteristics and responses.

Design a C++ program using polymorphism that includes:

A base class named Threat that contains:

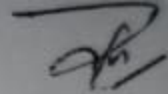
- A virtual function analyzeThreat() to be overridden in derived classes.
- A virtual destructor.

Three derived classes:

- Malware, which overrides analyzeThreat() to display a message like "Analyzing malware behavior signatures..."
- Phishing, which overrides analyzeThreat() to display a message like "Scanning suspicious URLs and email patterns..."
- Ransomware, which overrides analyzeThreat() to display a message like "Detecting file encryption activities..."

In your main() function:

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- 
- | | |
|--|--|
| <ul style="list-style-type: none">• Create an array of base class pointers (Threat*) pointing to objects of different derived classes.• Call the analyzeThreat() method polymorphically on each object. | |
|--|--|

***** End of Question Paper *****